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Article : Development of Performance Monitoring Dashboard for Product Packaging Manufacturer by using Waterfall Methodology

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1.0 Introduction

This report presents an interpretation of the case study of PT. MPM, a product packaging manufacturer for which improved monitoring and evaluation of operational performance were required. In the article, the development of a Business Intelligence (BI) dashboard is described as a response to the need for structured performance monitoring in the organization. The purpose of this tutorial report is to analyse the article from a system analysis and design perspective. The main problem, proposed solution, user requirements, Entity Relationship Diagram (ERD), use case diagram, data/system architecture, and multidimensional star schema are identified and discussed based on the case study.

2.0 Case Study Background

PT. MPM is described as a regional product packaging manufacturer in Indonesia. The company manufactures packaging products through both mass production and customized job-order production. The forecasts are received from customers in annual, quarterly and monthly format, and these forecasts are checked in conjunction with the historical sales figures, inventory availability and production planning information.

The article mentions that it was intended that the efficiency of the organization would be improved, but the necessary system for measuring and monitoring the work of the main business processes was not presented. Thus, it was suggested that a BI solution could be devised and presented as two dashboards: Sales Performance Dashboard and Production Performance Dashboard.

Dashboard	Main Purpose	Example Information Displayed
Sales Performance Dashboard	Sales performance is monitored and forecast values are compared with actual values.	Revenue, average revenue, total sales, forecast revenue, actual revenue, customer performance, and product category performance.
Production Performance Dashboard	Production performance and machine effectiveness are monitored.	OEE, availability, performance, quality, planned production, actual production, and production by machine.

Table 2.1: Overview of Business Intelligence Dashboards Developed for PT. MPM

3.0 Main Problem

As stated in the article, PT. MPM lacked an appropriate application or system for measuring and monitoring the performance of its core processes, causing top management to make decisions less effectively. MPM's core business processes. The data available on sales, production, inventory and ERP related reports were available, but not integrated, structured or visualized in an adequate manner for the managerial decision making process.

Consequently, the top management was not given performance information in an organized and central dashboard. Efficient comparison of planned and actual values was not possible and key performance indicators (KPIs) were not easily monitored. As a result, business information that was relevant was not made available in a timely and accessible manner, and therefore had an impact on decision making.

Problem Area	Explanation
Lack of a performance monitoring system	A suitable application or system for tracking sales and production KPIs was not available.
Scattered operational data	Data were distributed across sales, manufacturing, inventory, ERP, and reports, and were not integrated for analysis.
Manual and inefficient analysis	Data had to be reviewed and adjusted manually before it could be used for decision-making.
Weak decision support	Clear and current KPI information could not be accessed easily by top management.

Table 3.1: Summary of Main Problems Identified in PT. MPM

4.0 Main Solution

The key solution suggested in the article is the development of a Business Intelligence dashboard system. The data is gathered, cleansed and transformed, loaded into a data warehouse, prepared for analysis and presented in a dashboard to provide useful management information from raw operational reports.

The system has been developed using the Waterfall methodology which includes analysis, requirement, design, development, testing and implementation stages. This methodology was deemed appropriate as the requirements of the system were well-defined and the development process was sequential. Google Sheets was used as the data warehouse platform, while Google Data Studio was used for dashboard development and visualization.

Waterfall Phase	Application in the Case Study
Analysis	The business problem was identified, literature was reviewed, and data were gathered through interviews, observations, and discussions.
Requirement	Data were analysed and system requirements for sales and production performance monitoring were defined.
Design	The data warehouse was modelled and dashboard visualization designs were prepared and validated by management.
Development	ETL was performed, data were processed, and dashboards were developed using Google Data Studio.
Testing	Alpha and beta testing were conducted to evaluate functionality and user acceptance.
Implementation	The dashboard was implemented after the evaluation results were considered satisfactory.

Table 4.1: Application of Waterfall Methodology in the BI Dashboard Development

5.0 User Stories

User stories are used to describe the system requirements from the perspective of the intended users. Based on the article, the main users are represented by top management, the data analyst, the sales department, the manufacturing department, and the inventory and warehouse department.

User / Actor	Requirement Statement	Expected Benefit
Top Management	Access to sales and production dashboards is required so that business decisions can be supported by performance information.	Decision-making and performance monitoring can be improved.
Data Analyst	Data must be collected, cleaned, transformed, and uploaded so that accurate dashboard information can be displayed.	Reliable and consistent dashboard data can be maintained.
Sales Department	Sales data, including revenue, forecast sales, actual sales, product category, and customer information, are to be provided.	Sales performance can be analysed clearly.
Manufacturing Department	Production data, including planned production, actual production, runtime, stop time, and machine data, are to be provided.	Production performance can be tracked more effectively.
Inventory & Warehouse	Product stock data are to be updated so that stock availability can be reflected in the system.	Inventory visibility can be improved.

Table 5.1: User Requirements and Expected Benefits of the BI Dashboard System

The main user story can be stated as follows: As a top management user, access to an interactive BI dashboard is required so that sales and production performance can be monitored, KPIs can be reviewed, planned and actual results can be compared, and business decisions can be supported by accurate information.

6.0 USE CASE DIAGRAM

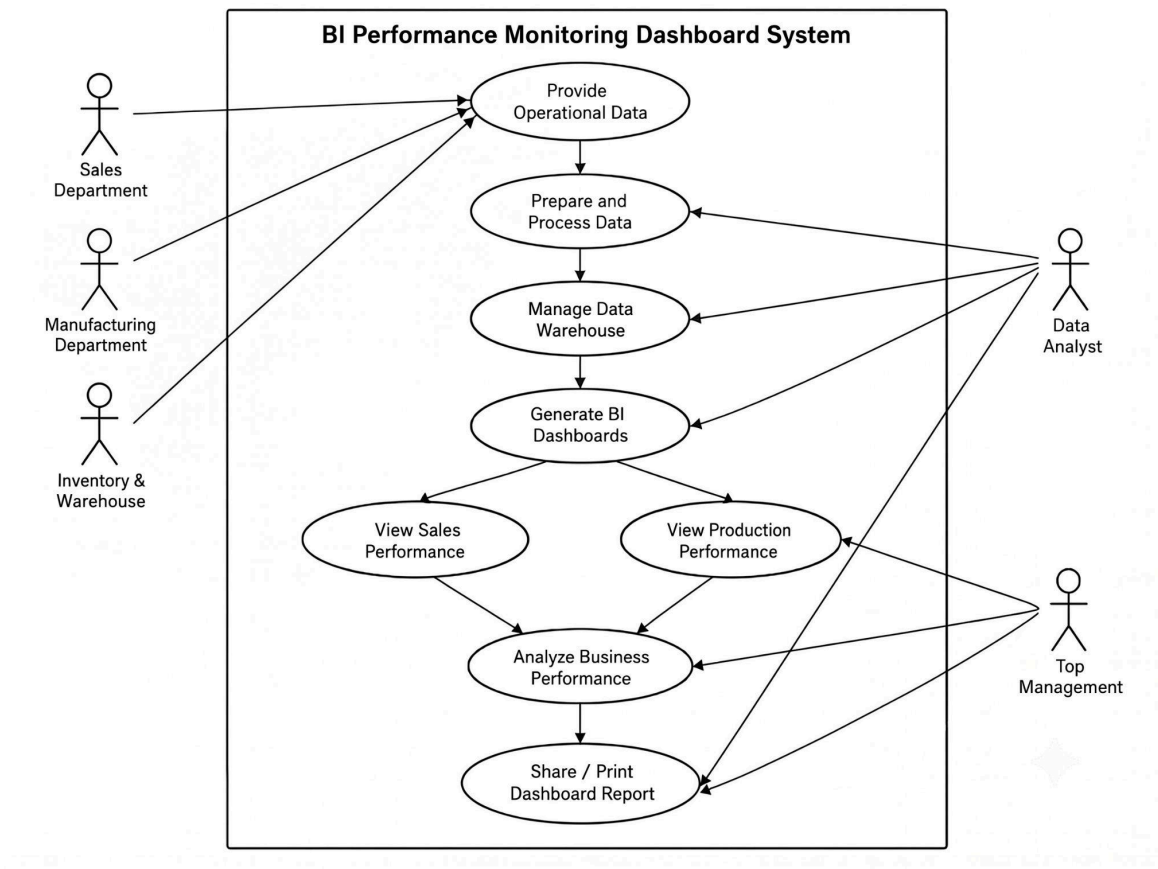


Figure 6.1: Use Case Diagram of the BI Performance Monitoring Dashboard System

Figure 6.1 presents the use case diagram of the BI Performance Monitoring Dashboard System for PT. MPM. The main interactions between the system and the users involved in sales and production performance monitoring are illustrated. The system boundary represents the BI dashboard system, while the actors outside the boundary represent the departments and users that are involved in data provision, data processing, dashboard usage, and reporting.

The left-side actors consist of the Sales Department, Manufacturing Department, and Inventory & Warehouse. These departments are represented as the main data providers for the system. Sales-related data are provided by the Sales Department, production-related data are provided by the Manufacturing Department, and stock or inventory data are provided by the Inventory & Warehouse department. These operational data are required to ensure that the dashboard can be generated based on relevant and useful business information.

The main system process is shown in the middle of the diagram. Operational data are first provided to the system. The data are then prepared and processed before being stored and managed in the data warehouse. After the data have been cleaned, transformed, and organized, BI dashboards are generated. The generated dashboards are then used to present sales performance and production performance. These performance views are further used to support business performance analysis and report preparation.

The right-side actors consist of the Data Analyst and Top Management. Data preparation, data processing, data warehouse management, dashboard generation, and report sharing are carried out by the Data Analyst. The dashboards are used by Top Management to view sales performance, view production performance, analyze business performance, and obtain dashboard reports for decision-making purposes.

Overall, the use case diagram shows that the BI Performance Monitoring Dashboard System is designed to connect operational departments with management users. Data from different departments are collected, processed, stored, and visualized through the dashboard system. Through this process, sales and production performance can be monitored more effectively, and management decision-making can be supported by clearer performance information.

7.0 ERD DIAGRAM

The ERD is proposed based on the operational data required by the BI dashboard. The proposed ERD is focused on entities that are needed to support sales performance analysis, production performance analysis, inventory monitoring, and dashboard reporting, as shown in Figure 7.1.

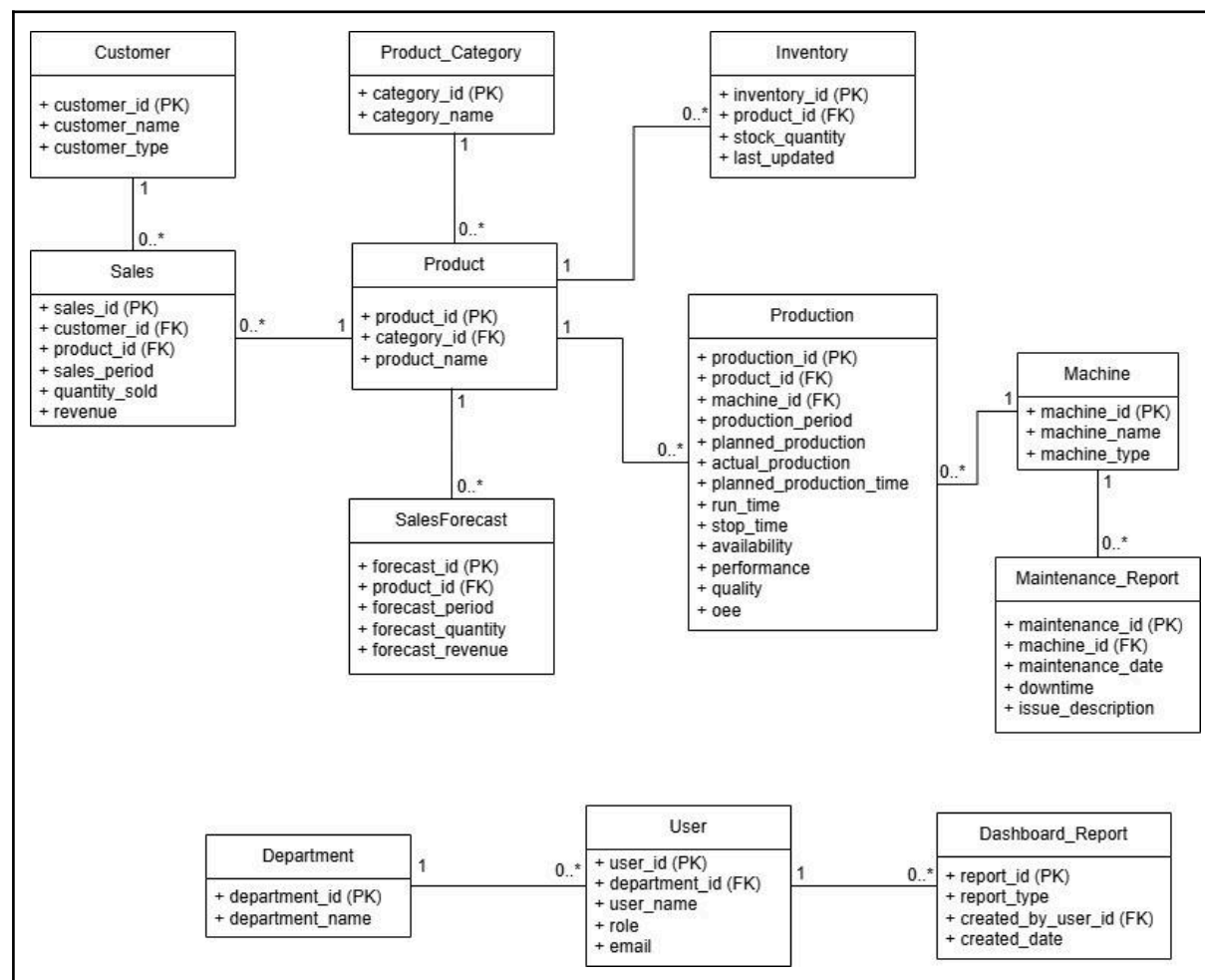


Figure 7.1 Proposed ERD using UML-style notation

7.1 Entity Description

Entity	Description
Customer	Customer information related to product sales is stored.
Product_Category	Product category information used for product classification is stored.

Product	Product details, including product ID, product name, and category information, are stored.
Sales	Sales transaction data, including customer, product, sales period, quantity sold, and revenue, are recorded.
SalesForecast	Forecasted sales information, including forecast period, forecast quantity, and forecast revenue, is stored.
Inventory	Product inventory information, including stock quantity and the latest update date, is maintained.
Production	Production performance data, including planned production, actual production, production time, run time, stop time, availability, performance, quality, and OEE, are recorded.
Machine	Information regarding machines used in the production process is stored.
Maintenance_Report	Machine maintenance records, including maintenance date, downtime, and issue description, are maintained.
Department	Department information associated with the BI dashboard system is stored.
User	User information, including role, email, and department affiliation, is maintained.
Dashboard_Report	Dashboard report information, including report type, creator, and creation date, is stored.

Table 7.1 Description of each entity

8.0 DESIGN DATA ARCHITECTURE

Figure 8.1 presents the proposed Data Architecture of the BI Performance Monitoring Dashboard System for PT. MPM. Operational data are obtained from various business sources, including the ERP system, sales reports, production reports, inventory data, product data, and maintenance reports. The data are first collected and integrated into a unified structure before undergoing processing activities such as data cleaning, data transformation, KPI calculation, and data validation. The processed data are then stored in a centralized data warehouse implemented using Google Sheets. Google Data Studio is subsequently used to generate Sales Performance and Production Performance dashboards that support KPI monitoring and trend analysis. The dashboards are accessed by Top Management and Data Analysts for performance monitoring, reporting, and decision-making.

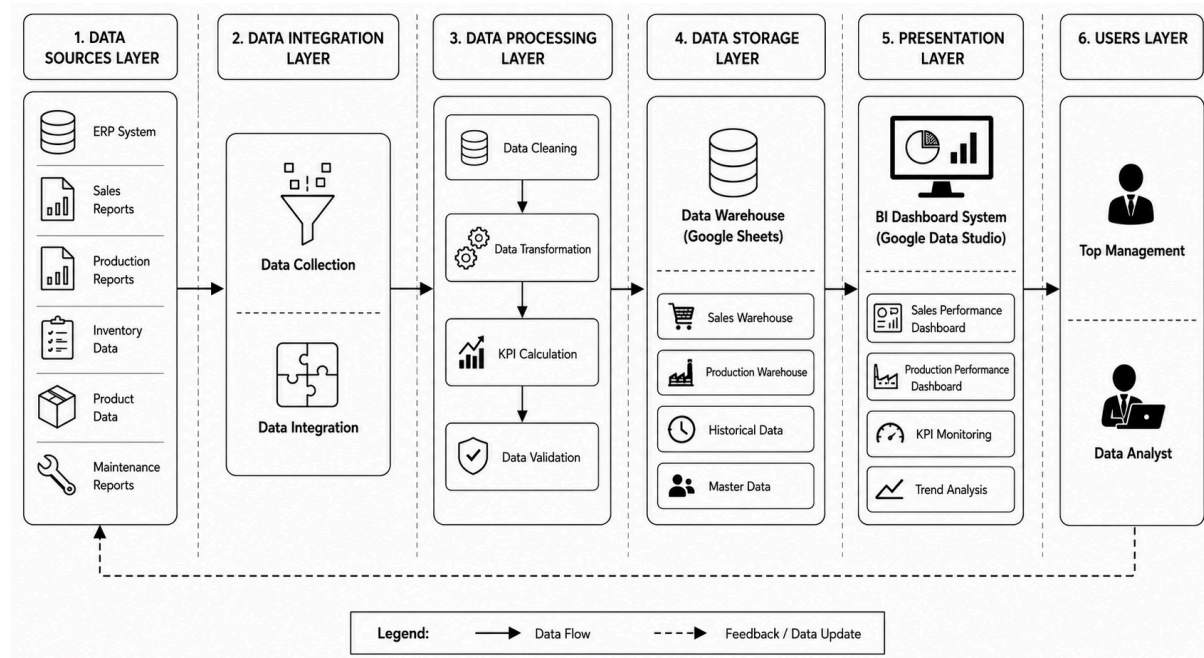


Figure 8.1 Proposed Data Architecture

8.1 Layers Description

Layer	Main Components	Description
Data Sources Layer	ERP System, Sales Reports, Production Reports, Inventory Data, Product Data, Maintenance Reports	Provides operational data from different business functions that serve as the primary source of information for analysis.

Data Integration Layer	Data Collection, Data Integration	Collects data from multiple sources and integrates them into a unified structure for further processing.
Data Processing Layer	Data Cleaning, Data Transformation, KPI Calculation, Data Validation	Processes and prepares data by improving data quality, standardizing formats, generating KPI metrics, and validating information accuracy.
Data Storage Layer	Data Warehouse (Google Sheets), Sales Warehouse, Production Warehouse, Historical Data, Master Data	Stores integrated and structured business data in a centralized repository to support reporting and analytical activities.
Presentation Layer	BI Dashboard System (Google Data Studio), Sales Performance Dashboard, Production Performance Dashboard, KPI Monitoring, Trend Analysis	Presents business information through dashboards and visual reports to support performance monitoring and analysis.
Users Layer	Top Management, Data Analyst	Represents the primary users of the system who utilize dashboard information for monitoring, reporting, analysis, and decision-making.

Table 8.1 Description of each layer

9.0 DESIGN MULTIDIMENSIONAL

A multidimensional model was designed to support the Business Intelligence (BI) Performance Monitoring Dashboard for PT. MPM. The model organizes operational data into fact tables and dimension tables to facilitate analytical processing, KPI monitoring, and management reporting. Based on the requirements identified in the case study, two star schemas were developed to support the Sales Performance Dashboard and Production Performance Dashboard. The star schema structure simplifies data analysis and improves query performance by separating measurable business data from descriptive business information.

9.1 SALES STAR SCHEMA

The Sales Star Schema is designed to support sales performance analysis. The Fact_Sales table stores measurable business data such as quantity sold, revenue, forecast quantity, and forecast revenue. The dimension tables provide descriptive information about customers, products, and product categories. This schema enables management to analyse sales performance, customer contribution, product category performance, and forecast accuracy.

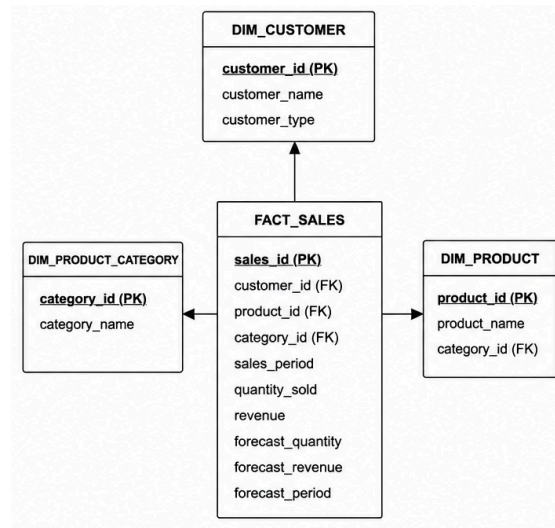


Figure 9.1 Sales Star Schema

Table Name	Table Type	Description
Fact_Sales	Fact Table	Stores sales transactions, quantity sold, revenue, forecast quantity, and forecast revenue.
Dim_Customer	Dimension Table	Stores customer information including customer name and customer type.
Dim_Product	Dimension Table	Stores product information used in sales transactions.
Dim_Product_Category	Dimension Table	Stores product category information for product classification and analysis.

Table 9.1 Description of Sales Star Schema

9.2 PRODUCTION STAR SCHEMA

The Production Star Schema is designed to support production performance monitoring. The Fact_Production table stores production KPIs such as planned production, actual production, availability, performance, quality, and Overall Equipment Effectiveness (OEE). The dimension tables provide descriptive information about products, product categories, and machines. This schema allows management to evaluate machine performance, monitor production efficiency, and compare planned production against actual production results.

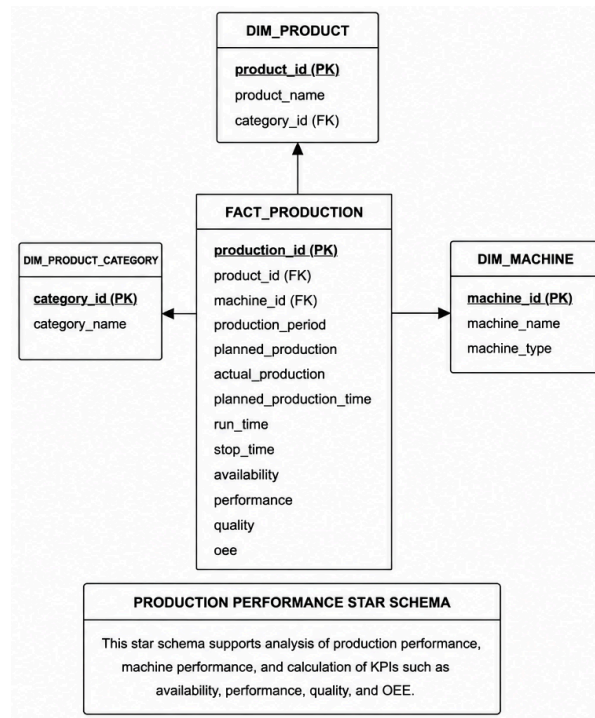


Figure 9.2 Production Star Schema

Table Name	Table Type	Description
Fact_Production	Fact Table	Stores production KPIs and performance measurements including OEE, availability, performance, and quality.
Dim_Product	Dimension Table	Stores product information related to production activities.
Dim_Product_Category	Dimension Table	Stores product category information for production analysis.

Dim_Machine	Dimension Table	Stores machine information used to analyse machine performance and production efficiency.
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Table 9.2 Description of Production Star Schema

The multidimensional model consists of two star schemas that support the Sales Performance Dashboard and Production Performance Dashboard. The Sales Star Schema focuses on sales analysis through customer, product, and product category dimensions, while the Production Star Schema focuses on production monitoring through product and machine dimensions. These schemas enable efficient KPI analysis, support management decision-making, and provide the foundation for the Business Intelligence dashboard system.